



COMMERCIAL LEO
DEVELOPMENT
PROGRAM

Joint Safety Reviews

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Joint Safety Reviews: Understanding Their Purpose and Importance



- NASA utilizes joint boards when safety responsibilities overlap to ensure review of information and a common understanding of safety risk across all programs
- This approach is applied in various contexts including:
 - **Shuttle/ISS, ISS/CCP, ISS/EHP, Orion/Gateway/HLS:** these programs have used or currently use joint panels or boards where applicable
 - **Board Responsibilities:** Each board reviews and concurs with acceptability of the safety risks associated with their specific area of responsibility
 - **Operational Framework:** Boards operate under their respective charters and processes, ensuring structured and consistent evaluations
 - **Documentation:** Some boards will create a Memorandum of Understanding (MOU) or Memorandum of Agreement (MUA) to document agreements and ensure clarity in roles and responsibilities
- *The structured approach ensures that all programs operate from a unified perspective, enhancing overall safety and compliance*



Multi-Panel/Board Reviews: conducted by various panels/boards, such as the CLDP Safety Review Board (CSRB), Commercial Crew Program Safety and Technical Review Board (CCP STRB), and the Commercial Provider Panel/Board. For example, integrated reviews may cover destinations, crew and cargo visiting vehicles (VV), payloads and cargo operations.

This structured approach ensures comprehensive and thorough safety reviews, leveraging the expertise of multiple panels and boards to maintain high safety standards across all operations.



- **Leadership of these reviews will be determined by the role of the Commercial Provider, CSRB, and/or CCP STRB.**
- Nominally led by NASA with participation of the Commercial Provider's panel/board
 - CSRB leads Integrated Destination reviews
 - CCP STRB leads reviews for activities outside the AE
 - CSRB and CCP STRB co-chair Visiting Vehicle integrated operations
 - Commercial Provider may lead if they have the established processes and infrastructure



Joint Safety Review Scope Summary

Safety Review		Board(s)/Panel	Objective/Criteria
Integrated Destination/Integrated Ops		Joint CSRB/CCP STRB/Commercial Provider Panel	Integrated Destination (Destination, crew and cargo visiting vehicles, utilization, payloads, cargo and core systems associated with USG crew habitation and ops), MWF (support USG crew) and/or MNWF (risk to crew), identification and assessment of emergency response actions Integrated HA submitted by Commercial Destination Provider
Destination and Launch Vehicle, Mission and Ground Systems		Joint CCP STRB/Commercial Provider Panel	Destination mated with launch vehicle, mission and ground system activities/operations related to launch and return HA submitted by Commercial Destination Provider
Crew Transportation System (Crew Visiting Vehicle and Launch Vehicle, Mission and Ground Systems)		Joint CCP STRB/Commercial Provider Panel	Full crew visiting vehicle and systems review, IHA (VV integrated ops), assess emergency response actions (USG crew survivability/aborts), return HA submitted by Transportation System Provider
Cargo Transportation System (Cargo Visiting Vehicle, Launch Vehicle and Mission and Ground Systems)		Joint Commercial Provider Panel/CCP STRB	Full visiting vehicle and systems review, integrated cargo safety review, IHA (VV integrated ops), disposal HA submitted by Transportation System Provider
Cargo		Joint CSRB/ Commercial Provider Panel Review	Limited to cargo the USG crew will interact with, includes impacts to common environment and all systems, personnel, and equipment that is controlling the hazard HA submitted by Cargo Provider
Utilization and Payloads	NASA Sponsored	CSRB Leading Panel	Limited to payloads the USG crew will interact with, includes impacts to common environment and all systems, personnel, and equipment that is controlling the hazard HA provided by Payload Provider, Payload Facility Owner, or Destination Provider
	Non-NASA Sponsored	Joint CSRB/Payload Provider Panel	



Partner Safety Review Board (SRB):

The Partner determines the composition of its safety board.

CLDP Safety Review Board (CSRB):

Chaired by CLDP.

Follows the CLDP-P-4007 process, CLDP Safety Review Board Charter.

The CSRB includes representatives from:

CLD Systems Engineering & Integration (SE&I)

CLDP Vehicle Office

CLDP Operations and Training Office

Safety and Mission Assurance Technical Authority (S&MA TA)

Engineering Technical Authority (TA)

Health and Medical Technical Authority (TA)

Flight Operations Directorate (FOD)

CCP Safety Technical Review Board (STRB):

Chaired by CCP.

Follows the CCT-P-4003 process, Commercial Crew Program Safety Technical Review Board Charter and Process.

The CCP STRB includes representatives from:

CCP Spacecraft Systems Office

CCP Launch Vehicle Systems Office

CCP Ground and Mission Operations Office

Chief Safety Officer (CSO)

Chief Engineer

Chief Health and Medical Officer

Flight Operations Directorate (FOD)

All safety reviews follow CLDP-REQ-3102



- **Commercial Provider is responsible for the safety of the Destination and its occupants**
 - Provider develops and maintains hazard analysis for the end-to-end mission
 - Provider is responsible for the integrated safety of payloads and operations
- **Visiting Vehicle Provider is responsible for the safety of the Vehicle and its occupants**
 - Develops and maintains hazard analysis for the end-to-end mission
 - Coordinates with Destinations and Utilization providers to ensure safe delivery/return (and disposal) of crew and utilization
- **Payload Provider is responsible for the safety of their end item/hardware**
- **NASA is responsible for ensuring the safety and well being of the USG crew during their mission and has responsibility for supporting NASA payload safety**
 - Need to maintain insight and confidence in Partner activities
- **There may be periods of time between USG Crews (~6 months)**
 - Need streamlined insight approach and program processes to maintain cognizance of the readiness of the destination for future missions



Commercial Panel/Board Roles and Responsibilities



Responsibilities of the **Commercial Provider** include:

- Detailed safety requirements for CLD, Visiting Vehicle, subsystems, payloads and cargo
- Integrated safety requirements and IRD for end item
- Demonstrate understanding and compliance to applicable safety process and technical requirements
- Perform a comprehensive safety assessment
 - Ensure the hardware is safe throughout all mission phases and operations; including all interactions between the vehicle (launch, on-orbit, and return), Destination elements/modules, and other payloads
 - Develop and submit SDP with supporting HRs
 - Control all potential hazards and document in detailed HRs
 - Provide appropriate documentation to support the Phase-Level safety review
 - Closed loop traceability from hazard controls to requirement verifications and supporting documentation (including formal risk acceptance of non-compliances)
 - Coordinate with other Commercial Providers if hardware relies on a hazard control provided by other commercial equipment and/or module/vehicle



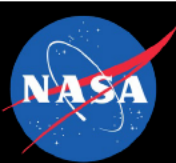
Responsibilities of the **Commercial Provider** include:

- Implementation of real-time status and processes
- Develop and maintain hazard analysis for the end-to-end mission – assess data required to support NASA’s human rating of US Government crewed missions
- Regular updates to safety analysis/supporting data to incorporate new safety findings, technologies, and methodologies
- Engage a wider range of stakeholders in the review and improvement process to gather diverse perspectives and expertise
- Provide technical and management resources necessary to ensure the overall safety and protection of flight and ground personnel, general public, flight/ground hardware, the environment, transportation systems and facilities through all phases of the applicable contract, including oversight of subcontracted efforts
- Defined chartered safety panel/board
- Defined approval/concurrence processes from the Commercial Provider’s PM
- Formal dissent process



Responsibilities of the **Commercial Provider** include:

- Commercial Providers will determine the composition of their panel/board – provide technical and management resources necessary to ensure overall safety of crew, flight hardware, transportation systems, destinations, etc. throughout all phases
- Have provisions in place to permit CLDP access to the prime Commercial Provider safety data, as well as Provider Subcontractor safety data
- Develop training modules or guidelines to help users understand and effectively apply the document's requirements and recommendations
- Set up configuration management of data, HRs, hazard controls implemented in training, flight rules and/or flight procedures, non-compliance/variance data, supporting data, and verifications
- Action capture and tracking
- Method to identify, track, and resolve errors/discrepancies in approved hazard reports
- Capture of minutes for safety reviews, WGs, and OSB meetings
- Maintain master schedule for payload, end item, integrated safety reviews



CSRB Roles and Responsibilities



Responsibilities of the **CSRB** include:

- Assists in the interpretation of safety requirements and provides recommendations for implementation
- Conducts safety reviews to assess submitted safety analyses, hazard reports, and safety non-compliances
- Evaluates hazard assessments resulting from modifications that either affect a safety-critical subsystem or create a hazard to the Destination, visiting vehicle, or crew
 - Reviews design and operations to ensure all hazards are identified and have their Risk appropriately described (Likelihood and Consequence)
- Verifies that all hazards are controlled in accordance with Program requirements (FT, DFMR, NCR/variance)
- Coordinates within the CLDP as well as with other NASA Program(s) boards, panels, and working groups as necessary for the assessment of Destination integrated safety topics
- Coordinates and negotiates the resolution of safety issues, including (as required) the formation of splinter groups/sub-panels to perform technical activities required to accomplish the objectives of the CLDP
- Reports the findings and recommendations of the CSRB, including issues which cannot be resolved within the board(s) and require elevation to CLD Program management
- Elevates to the CLDP Safety and Mission Assurance Control Board (CSMACB) and the CLDP Program Control Board (CPCB), those safety non-compliances above the delegated approval authority of the CSRB as defined in CLDP-REQ-3102



Joint Safety Review Completion

Completion of Phase III reviews is normally accomplished by obtaining approval of all critical and catastrophic HRs.

The Commercial Provider shall meet the following criteria for successful completion of a Phase III safety review:

- Safety review Action Items (AIs)/agreements associated with HRs are formally closed, if applicable.
- Final updates to non-compliance/variance rationale incorporated in HRs, if applicable.
- Panel/Board comments have been dispositioned. HA Owner/Provider disposition of comments are discussed at the phased safety review.
- If all activities associated with the safety review (other than open verifications) are completed, all HRs are signed by the Commercial Provider Panel delegated authority, CSRB Chair and/or CCP STRB Chair at the Phase III level.
- Final Safety Verification Tracking Log
- Final signed SDP submitted by HA Owner/Provider Project Manager or delegated authority after successful completion of the items above. Final safety assessment approval by the CSRB Chair and/or CCP STRB Chair, when applicable.
- Final signed Flight Safety Certificate